Project Design Phase-I
Solution Architecture

Date	16 February 2026
Team ID	LTVIP2026TMIDS63806
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Solution Architecture:

Heart Disease Analysis Dashboard

The platform follows a layered system architecture designed to efficiently collect, process, analyze, and visualize heart disease and patient health data.

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1. Data Providers & Sources

Sources:
- Hospitals and clinics
- Electronic Health Record (EHR) systems
- Diagnostic laboratories
- Public health datasets (e.g., government health portals)
- Wearable health monitoring devices (optional future scope)

Data Generated:
- Patient demographics (age, gender)
- Clinical metrics (cholesterol, blood pressure, heart rate, ECG results)
- Lifestyle indicators (smoking, exercise, diet)
- Diagnosis outcomes and risk classifications

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2. Data Ingestion Layer

Methods:
- CSV/Excel file uploads from hospital systems
- Database connections (MySQL, PostgreSQL)
- API integrations from EHR systems
- Scheduled automated data imports

Purpose:
To systematically collect structured and unstructured patient data into the analytics environment.

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3. Raw Data Storage Layer

Infrastructure:
- Secure cloud storage (AWS S3, Google Cloud Storage)
- Local hospital servers
- On-premise data repositories

Purpose:
Ensures secure storage and availability of raw patient data for historical and longitudinal analysis while maintaining compliance with healthcare data policies.

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4. Data Processing & Transformation Layer (ETL)

Tools:
• Python (Pandas, NumPy)
• Tableau Prep
• ETL pipelines

Actions:
• Handle missing or inconsistent values
• Standardize medical formats (e.g., BP units, cholesterol levels)
• Categorize age groups and risk levels
• Aggregate data by demographics and clinical indicators
• Prepare analytics-ready datasets

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5. Data Storage & Query Layer

Technologies:
• PostgreSQL / MySQL
• Cloud-based data warehouse

Purpose:
Stores cleaned, structured patient data optimized for fast analytical queries and scalable access by BI tools.

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6. Analytics & Visualization Layer

Tools:
• Tableau Desktop
• Tableau Server / Tableau Public
• Power BI (optional)

Outputs:
• Demographic analysis dashboards (age & gender distribution)
• Risk factor correlation charts (cholesterol vs BP)
• High-risk patient identification views
• KPI cards (Total Patients, High-Risk %, Avg Cholesterol, Avg BP)
• Trend analysis for monitoring patient health patterns

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7. Monitoring & Alerts Layer (Optional / Advanced Scope)

Features:
• Threshold-based alerts for critical health indicators
• Automated weekly or monthly clinical reports
• Email notifications for high-risk patient detection
• Predictive risk scoring (future enhancement)

8. End Users & Stakeholders

Audience:
• Cardiologists
• General physicians
• Hospital administrators
• Public health researchers
• Healthcare policymakers

Outcomes:
• Early identification of high-risk patients
• Improved clinical decision-making
• Better preventive healthcare planning
• Enhanced reporting and patient outcome tracking
• Data-driven healthcare management

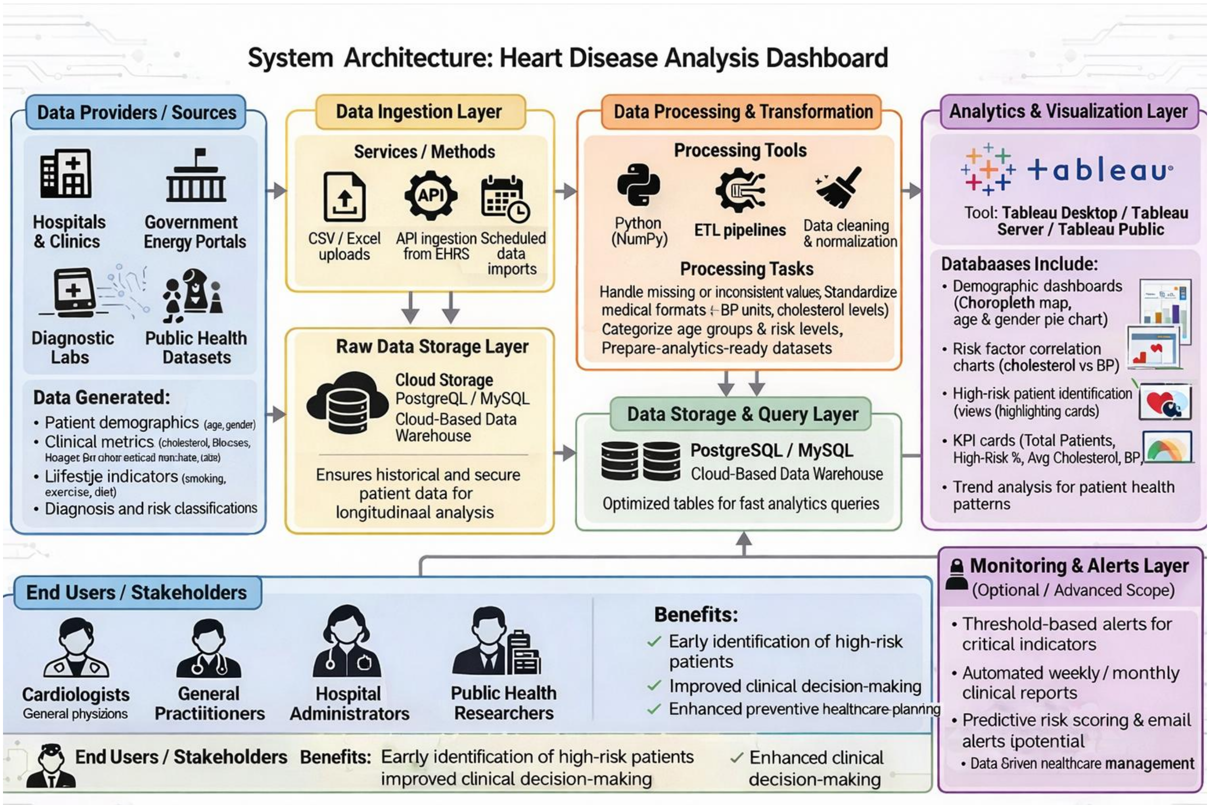

Fig:-System Architecture (Heart Disease Analysis)